Günter Blobel (1936–2018)

When Günter Blobel died on February 18th, the world lost an innovative thinker and superb experimentalist who ushered cell biology into the molecular age. He left new paradigms that continue to inform our understanding of how cells work, a legion of trainees who continue to transform science, and a legacy of accomplishments as a global citizen. Debonair and jovial, he had a ferocious passion for science as well as opera and architecture.

Blobel was awarded the Nobel Prize in Physiology or Medicine in 1999 "for the discovery that proteins have intrinsic signals that govern their transport and localization in the cell." In the mid-20th century, cell biology was undergoing a revolution. Elegant morphological and biochemical assays were characterizing cellular architecture, and a cataloging of "what was where" had identified the subcellular compartments, their contents, and their functions. Blobel's demonstration of discrete "topogenic sequences" in proteins that governed where they were in the cell, how they got there, and how they were folded provided the first mechanistic insight into how this architecture was built. Blobel's models of protein topogenesis provided critical insights into normal physiology and numerous diseases and helped launch emerging fields of biotechnology, such as the production of human proteins in other organisms.

The Nobel committee cited Blobel's "elegant biochemical experiments" achieved through years he spent in the cold room. For those who worked in Blobel's lab, it was clear, even after he no longer worked at the bench, that his passion was the pursuit of the experimental test. Blobel routinely held court in his office–with a carton of clementines always by his side, he analyzed data, discussing the pitfalls and promises of experiments, preferring to merrily conjure up images through gesticulation rather than writing on boards. The well-justified attention given to his experimental prowess did not overshadow his biological intuition. Blobel was renowned for expositing hypotheses, if one liked them, or reckless speculations, if one did not. His inspiration was often to ask, "How would I build such a thing?" But he judged the beauty of any hypothesis by its amenability to experimental validation. These hypotheses, which impacted many aspects of biology, seemed to emerge from Blobel fully formulated. Some emerged only after he returned from a trip to the opera, suggesting a potential source for the inspiration.

Blobel's approach may have had its roots in his childhood in a small town in Silesia in eastern Germany. In February of 1945, near the end of the World War II as his family was retreating from the advancing Russian armies, they entered the city of Dresden. Blobel, who was 8 years old, had never seen a city. He was enchanted by the many towers, especially the grand cupola of the Frauenkirche that rose above the city. They "were a magnificent sight even for the untrained eye of a child," he recalled, and not for the last time, he wondered, "How would I build such a thing?" "I was absolutely enchanted and always remained attached to the city because I saw it in a completely intact way." After leaving Dresden, the family stopped a few days at a relative's farm to the west of the city. At night, they heard the fleets of planes heading toward Dresden as the night sky turned red from fire. The young Blobel was angered by the loss of the magical city. "You just don't destroy beautiful things. You don't destroy Amsterdam; you don't destroy Venice, ever. These are holy places, not holy by the church but in terms of human creativity. They are ensembles that are so perfect that you don't touch them." Two months after the firebombing, Blobel was hit by the human expenses of war: his 19-year-old sister Ruth was killed while riding a train that was hit in an aerial bombing.

After the war, Blobel's family settled in Freiberg, Germany, where he developed a great love for theater and music. Each week he listened to Bach cantatas performed in the local church on an organ built by one of Bach's friends, Gottfried Silbermann. Blobel was taken with the great choral works and yearned to pay tribute to great music, even performing in Bach's Matthäus Passion.

Later, Blobel studied medicine, which both fascinated and frustrated him. He loved learning about biology but was irritated by medicine's descriptive, correlative nature—the inability to demonstrate cause and effect. His oldest brother, Hans, had followed in their father's footsteps and had become a veterinarian. He, too, was frustrated by the limitations of knowledge and switched to research. He had moved to the United States and become a professor of microbiology at the University of



Günter Blobel with his wife Laura Maioglio, whose restaurant Barbetta has hosted countless scientific debates.

Wisconsin in Madison. After two years of his medical internship, Blobel followed his brother to Madison to do his graduate work with Van R. Potter. There, Blobel quickly established what would be the major foci of his work for the next 55 years: protein targeting to cellular membranes and nuclear transport.

In Potter's lab, Blobel had two projects. The first aimed at achieving a high-yield purification of the nucleus. The second project examined the timing of protein translation and targeting of mRNA and proteins to membranes. At that time, key protein-target experiments were being done in the lab of George Palade at The Rockefeller University. Palade, along with Carl Redman, Phil Siekevitz, and David Sabatini, addressed a key question: do proteins target to and translocate across the membrane as they are synthesized or only after? They used puromycin to release nascent proteins from their biosynthetic ribosomes. This showed that the released peptides, no matter how short, were found inside of the membrane-bound organelles. They rightfully concluded that the proteins translocated as they were synthesized. Blobel moved to Palade's lab as a postdoc where he was thrilled to join colleagues deep in mechanistic discussions: Was the targeting information in the RNA or the protein? Was the RNA targeted to ribosomes on the membrane that are distinct from those that make soluble proteins? Blobel teamed up with Sabatini to show that soluble and membrane-bound ribosomes were interchangeable. Before publishing this result, they proposed "The Signal Hypothesis," in a brief three-page book chapter. The Signal Hypothesis rested on three assumptions: first, that protein synthesis began on free ribosomes; second, that the protein has a signal in its amino acid structure, a "signal sequence" that is responsible for targeting to membranes; and third, that there are proteins in the target membranes.

While the hypothesis was still speculative, it made very specific experimentally testable predictions. Soon, other labs published results consistent with the hypothesis. Specifically, the light chains of immunoglobulins that were translated in a cell-free system migrated slower on a gel than those secreted from the cell. This suggested that the secreted immunoglobin might be smaller, possibly the result of cleavage of the speculated signal sequence. Alas, the change in mobility on a gel was inconclusive. In the cell-free system, the slower mobility could be the result of a mis-start of translation, or the faster mobility of the secreted protein could be the action of a non-specific protease.

To demonstrate a causal relation, Blobel borrowed from the biochemists' playbook: he would fractionate each of the components and ask what must be reconstituted to achieve targeting to and transport across the membranes. This was much easier said than done as a reconstitution of a membrane-transport process had not been achieved. How would one know whether the protein actually crossed, and how would one determine what was a signal? Blobel started with the obvious components: ribosomes; a cytosolic translation mix; mRNA for secreted protein; and membranes from the endoplasmic reticulum, the entry point for the secretory pathway. The results were consistent and a failure. Blobel tried fractionating rat liver, then pigeon pancreas, which were in standard use in the lab. When those failed, one by one, he tried different organs and every different organism that was in use or available at the Rockefeller University. Nothing worked. It was unclear whether there was a flaw in the experimental setup, in his technique, or in the hypothesis.

Finally, the first positive signal came after years in the cold room. However, even then there was no "eureka moment." Blobel knew a signal could be the result of many unforeseen problems. Critical was the effort spent in controls: Did the nascent protein harvest with the membranes? Was it transported across the membranes? Was the protein protected from exogenously added protease, and was that protection really the consequence of being shielded inside of the membrane of the endoplasmic reticulum? The work was the critical reconstitution of a membrane-transport process. It stimulated both parallel work by many other groups on protein targeting to membranes as well as stimulated new directions, such as Jim Rothman's landmark reconstitution of vesicular transport. For Blobel, it was just the opening salvo in his study. In these papers, he speculated further on the role of the signal sequence, the cytosolic factors, and the protein-conducting channel, which he posited was the transit pathway for passage across the membrane.

The next few years were a flurry of activity. The Blobel lab demonstrated that if one mutated the postulated signal sequence, proteins no longer targeted. If one transplanted the signal sequence to cytosolic protein, it did target. The "Blobelites," as the lab members named themselves, repeated the approach of fractionation and reconstitution. Fractionation of the cytosol led to the identity of the cytosolic factor that chaperoned the nascent protein to the appropriate membrane: the signal recognition particle (SRP). Fractionation of the membrane led to the identification of the receptor for SRP. Their work established the signal sequence as a general mechanism for protein targeting to the mitochondrion, chloroplast, peroxisomes, and even to the bacterial plasma membrane.

Blobel proposed that the signal sequence for targeting was but a specific example of the more general signals for protein topogenesis. His lab demonstrated that signal sequences, together with SRP, were required for the targeting and orientation of transmembrane segments, such as the acetylcholine receptor and opsin. They expanded their work to examine the role of such sequences in the transport of proteins from one cell surface to another, such as used for transcellular transport of immunoglobulins. They used biophysical techniques to demonstrate the existence of Blobel's hypothesized protein-conducting channel and explored its interactions with both the ribosome and the signal sequence. An additional fractionation led to reconstitution of the protein-translocation machinery.

In Van Potter's lab, Blobel had started two projects. The second resulted in a beautiful paper published in *Science* on the purification of the nucleus of the cell. Blobel's lab continued the pursuit of the nucleus, leading to purification of the nuclear pore complex (NPC) and its components and of the nuclear lamina. The Blobelites continued the approach of fractionation and reconstitution, in this case in pursuit of nuclear transport. This led to the identification of cytosolic factors they called karyopherins, from the

Greek "to carry" for nuclear transport. These worked in an analogous fashion to SRP, recognizing the topological signals for nuclear localization or nuclear export. A gradient of RanGTP:RanGDP across the nuclear envelope provided the chemical potential gradient for transport.

Blobel always adopted whatever techniques were required to pursue the biological question: biochemistry, genetics, electrophysiology. In the last phase of his work on the NPC, he turned to structural biology. One-by-one the Blobelites picked off the proteins of the NPC and crystallized them. Blobel observed two different conformations of key proteins of a ring in the nuclear pore and proposed that these proteins underwent a conformation change similar to a *Mädchenfänger*, also known as a finger trap. Binding of the karyopherins anchored to filaments attached to these proteins triggered a massive conformational change that would produce a reversible narrowing and widening of the channel.

Blobel had the opportunity to repay one debt for his inspiration when he named the proposal "The Ring Cycle" in tribute to his favorite operas. In fact, Blobel had the opportunity to repay many of his debts in his later years. After he won the Nobel Prize, he donated his winnings, in memory of his sister Ruth, to his first inspiration for "How would I build it," rebuilding the Frauenkirche (that was destroyed in the fire-bombing of Dresden) and building a replacement for the old Jewish synagogue (that was destroyed by the Nazis on Krystallnacht).

Though the science community has lost Günter Blobel, it will be impacted by his contributions for a long time to come. His work on fractionation and reconstitution brought cell biology into the molecular age. His rigor in his experimental pursuit established assays that continue to be the model decades later. Blobel loved his speculations but always reminded his trainees of Aldous Huxley's adage that "beautiful ideas die from ugly facts." One must be prepared to drop one's models, Blobel said. He is often remembered for his ideas, the Signal Hypothesis, protein topogenesis, gene gating, the protein-conducting channel, the Ring Cycle. The Nobel committee was right when it honored him for his "elegant biochemical experiments." However, acclaim should also go to his speculations to explain cellular function that have provided the motivation for half-a-century's worth of experiments in cell biology.

Sanford M.Simon[1,*]
[1]Rockefeller University, New York, NY 10065, USA
*Correspondence: simon@rockefeller.edu
https://doi.org/10.1016/j.cell.2018.03.047